

Can Language Models Perform Robust Reasoning in Chain-of-thought Prompting with Noisy Rationales?

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Abstract

This paper investigates an under-explored challenge in large language models (LLMs): chain-of-thought prompting with *noisy rationales*, which include irrelevant or inaccurate reasoning thoughts within examples used for in-context learning. We construct NoRa dataset that is tailored to evaluate the robustness of reasoning in the presence of noisy rationales. Our findings on NoRa dataset reveal a prevalent vulnerability to such noise among current LLMs, with existing robust methods like self-correction and self-consistency showing limited efficacy. Notably, compared to prompting with clean rationales, GPT-3.5 drops by 1.4%-19.8% in accuracy with irrelevant thoughts and more drastically by 2.2%-40.4% with inaccurate thoughts.

Addressing this challenge necessitates external supervision that should be accessible in practice. Here, we propose the method of contrastive denoising with noisy chain-of-thought (CD-CoT). It enhances LLMs' denoising-reasoning capabilities by contrasting noisy rationales with *only one clean rationale*, which can be the minimal requirement for denoising-purpose prompting. This method follows a principle of exploration and exploitation: (1) rephrasing and selecting rationales in the input space to achieve explicit denoising and (2) exploring diverse reasoning paths and voting on answers in the output space. Empirically, CD-CoT demonstrates an average improvement of 17.8% in accuracy over the base model and shows significantly stronger denoising capabilities than baseline methods. The source code is publicly available at: <https://github.com/tmlr-group/NoisyRationales>.

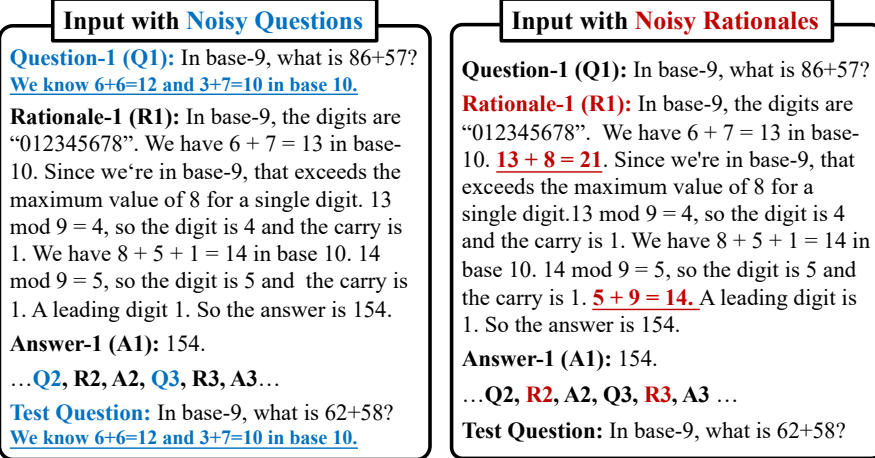


Figure 1: Exemplars of noisy questions [68] and *noisy rationales* (our new research problem). Each input includes three prompting examples and one test question. Notably, the test question asks about base-9 calculation, while the misleading base-10 information is given in noisy questions or rationales.

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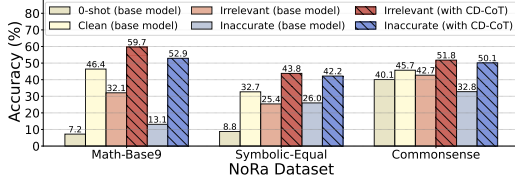


Figure 2: Results of GPT-3.5 with 0-shot, 3-shot clean rationales, and 3-shot noisy rationales: Both inaccurate and irrelevant rationales degenerate performance significantly, while the proposed CD-CoT improves robustness against noisy rationales.

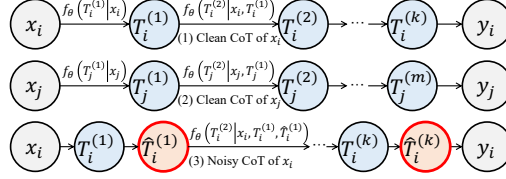


Figure 3: Chain modeling of the noisy rationale problem: Recovering chain (3) from chain (1) with the guidance of chain (2). From question x_i to answer y_i , the rationale of chain (3) includes clean thoughts $T_i^{(j)}$ and noisy thoughts $\hat{T}_i^{(j)}$.

1 Introduction

In-context learning (ICL) is a common approach in large language models (LLMs), enabling models to extrapolate from a few examples and adapt without fine-tuning [4, 84, 16]. However, ICL’s efficacy is closely tied to the quality and clarity of the prompting examples, particularly in the prevailing chain-of-thought (CoT) strategy that provides rationales, *i.e.*, intermediate reasoning steps to solve a question [85]. Recent research has shown that LLMs struggle with noisy questions: they are easily distracted by irrelevant context and exhibit instability with slight input modifications [68, 78, 107].

Notably, this work shifts focus from the well-studied noisy questions (Noisy-Q) problem to the under-explored *noisy rationales* (Noisy-R) problem, wherein *factually inaccurate or irrelevant reasoning steps* are paired with valid question-answer examples, as illustrated in Fig. 1. Here, the emphasis on Noisy-R is due to its practical challenges, with examples drawn from diverse sources such as crowd-sourced platforms, dialogue systems, and machine-generated data² [25, 45, 73, 2, 77, 48]. However, the robustness of LLMs against Noisy-R remains unknown. A new benchmarking dataset is needed to conduct a systematic evaluation of current LLMs and verify the corresponding countermeasures.

In this work, we first construct the NoRa (**Noisy R**ationales) dataset, a comprehensive testbed to evaluate the robustness of LLM reasoning against noisy rationales across various reasoning domains (in Sec. 3). The NoRa contains a total of 26391 questions, covering three types of reasoning tasks: mathematical, symbolic, and commonsense. We uniformly formalize the generation of noisy rationales by inserting irrelevant or inaccurate thoughts, controlling reasoning difficulty through noise ratios, and guaranteeing the overall prompting correctness without modifying the question or answer.

With the NoRa dataset, we evaluate several LLMs and reveal that all of them are *intrinsically vulnerable* to noisy rationales (in Sec. 4). For example, compared to prompting with clean rationales, GPT-3.5 exhibits an average 3.0% - 33.3% decrease in accuracy with noisy rationales, as in Fig. 2. Besides, only limited improvements are achieved with existing robust methods based on the model’s intrinsic denoising ability, *e.g.*, self-consistency [83] and self-denoise [102]. We show that Noisy-R is much more challenging than Noisy-Q, requiring context-specific knowledge to guide the denoising.

To solve this, we propose to rectify the rationales with *only one clean CoT demonstration* that can be the most attainable supervision in practice (in Sec. 5). We assume that LLMs can rectify rationales by *contrasting* a noisy rationale with a clean one, as in Fig. 3. Guided by this principle, we design the framework of **Contrastive Denoising with noisy CoT (CD-CoT)** with four steps: rationale rephrasing, rationale selecting, rationale exploring, and answer voting. Technically, the first two steps aim to achieve explicit denoising, while the last two steps are for diverse reasoning paths. Empirically, CD-CoT achieves an average improvement of 17.8% in accuracy *w.r.t.* the base model (refer to Tab. 8). Notably, it presents much stronger denoising power than baselines in rectifying the rationales.

Contributions. To our best knowledge, we are the *first* to investigate the problem of noisy rationales.

- We formalize the under-explored noisy rationale problem in the prevailing chain-of-thought prompting and construct the NoRa dataset to benchmark the robustness of LLMs against noisy rationales (Sec. 3).
- We systematically evaluate LLMs with NoRa dataset and extract several insightful observations, *e.g.*, the unsatisfactory robustness and limited denoising power of LLMs under noisy rationales (Sec. 4).
- We propose to rectify the noisy rationales with only one clean CoT demonstration, design a simple yet effective method, CD-CoT, and verify its effectiveness through comprehensive experiments (Sec. 5).

²The sources of noisy rationales are discussed in Appendix C with extensive real-world examples.

2 Related Work

Limitations of in-context learning (ICL). Though effective, ICL suffers from the susceptibility to manual generation and selection of demonstrations (examples), where the ultimate performance is closely tied to the demonstrations’ quality and clarity. Recent investigations on *noisy questions* have shown that (i) LLMs can be distracted by irrelevant or adversarial context, as they are designed to pay close attention to the context provided in the prompt [32, 58, 68, 78] and (ii) LLM reasoning is unstable, namely, small modifications to the prompt could potentially cause large variations in the model’s output [102, 107]. Besides, another line of research regarding *noisy answers* [42, 18] justifies the feasibility of misleading an LLM to agree factual errors such as "1+1=3" in base-10 calculation.

Countermeasures. Two intrinsic traits of LLMs are desirable for addressing the above limitations:

- *Self-correction*, wherein LLMs attempt to correct their initial responses based solely on their inherent capabilities without external feedback, *e.g.*, by refining prompts through iterative corrections of responses or question trajectories [91, 89]. Although LLMs can learn to ignore irrelevant information by examples or instructions [68], they are proved to be still struggling to correct their responses without external feedback, and at times, their performance might even degrade after self-correction [29, 81].
- *Self-consistency* aims to obtain a consistent answer against input perturbations. This is achieved by generating multiple responses via randomized smoothing on input questions [98] or diverse paths for answering one question [83] followed by the answer aggregation. This strategy brings improvements with extra costs for repeated reasoning. Moreover, it cannot explicitly rectify questions or rationales.

Noisy rationales, as the research focus of this work, mainly originates from (1) the inherent imperfections, inconsistencies, and inaccuracies of humans’ cognitive processes [53, 10] and (2) the diversity, unpredictability, and hallucination of the LLMs’ generative mechanisms [103, 30, 101]. A detailed literature review and discussion of noise rationales are in Appendix B and C, respectively.

3 The NoRa Dataset

In this section, we introduce the NoRa (**Noisy Rationales**) dataset for benchmarking the robustness against noisy rationales. NoRa consists of 26391 questions and 5 subsets, covering mathematical, symbolic, and commonsense reasoning tasks, where ICL and CoT demonstrations play a crucial role.

3.1 Definition of Noisy Rationales

We start by formalizing the ICL and CoT demonstrations. Given a test question x_{test} and an LLM f_θ , one expects to get the correct answer y_{test} as $f_\theta(x_{\text{test}}) \mapsto y_{\text{test}}$. This zero-shot manner cannot guarantee effectiveness, especially when encountering unfamiliar contexts or scenarios. To boost effectiveness, the ICL techniques prompt the LLM with a few examples $S_n = \{(x_i, y_i)\}_{i=1}^n$ collected in the current context, each composed of a question x_i and answer y_i , and then construct the new input x_{ICL} as

$$x_{\text{ICL}} = [S_n, x_{\text{test}}] = [x_1, y_1, \dots, x_n, y_n, x_{\text{test}}]. \quad (1)$$

The guidance by S_n makes $f_\theta(x_{\text{ICL}}) \mapsto y_{\text{test}}$ much easier than $f_\theta(x_{\text{test}}) \mapsto y_{\text{test}}$. Then, the CoT further refines x_{ICL} by constructing the step-by-step rationale $\mathcal{T}_i$, consisting of several thoughts $T_i^{(j)}$, namely,

$$x_{\text{CoT}} = [x_1, \mathcal{T}_1, y_1, \dots, x_n, \mathcal{T}_n, y_n, x_{\text{test}}], \text{ where } \mathcal{T}_i = [T_i^{(1)}, T_i^{(2)}, T_i^{(3)}, \dots, T_i^{(k)}]. \quad (2)$$

However, as aforementioned, the thoughts in CoT (Eqn. 2) can be noisy in practice. This noise can be attributed to (1) *irrelevant thoughts*, which are irrelevant but correct, or (2) *inaccurate thoughts*, which are relevant but factually wrong. Here, we *uniformly* formalize these two kinds of noise as

$$\hat{\mathcal{T}}_i = [T_i^{(1)}, \hat{T}_i^{(1)}, T_i^{(2)}, \hat{T}_i^{(2)}, \dots, T_i^{(k)}, \hat{T}_i^{(k)}], \quad (3)$$

where $\hat{T}_i^{(j)}$ represents a noisy thought (irrelevant or inaccurate) that is coherent with the previous clean thought $T_i^{(j)}$ (relevant and correct). The following introduces the definition of noisy thoughts.

Irrelevant thoughts refer to incorporating irrelevant information unhelpful for solving the question, *e.g.*, discussing the genetic overlap of siblings when the task is to deduce family roles in relationship reasoning. Redundant information may be introduced by the LLM’s diverse response generation or